The Natural Philosophy of Universal Matter

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Abstract

This paper explores the formation of a hydrogen atom. Employing the philosophy of reductionism, we examine the subatomic world and delve deeply to establish the foundations of matter. Next, we discuss the rationale for focusing on this level of matter and describe its key characteristics. We then examine these concepts within the framework of the Standard Model of particle physics. Using this understanding, we identify and define the natural physical processes required to form an atom. Finally, we analyze the interplay between the strong and electromagnetic forces and outline the steps involved in atomic formation.

1 Introduction

John Polkinghorne, in his article *Reductionism*, states that "a reductionist believes that a complex system is nothing but the sum of its parts."[1] Elana Popan, in her article *Reductionism (philosophy)*, similarly writes that "everything that exists is a collection or combination of entities of a simpler or more basic kind."[2] As we examine the universe, some of these simpler parts are fairly well understood, including objects such as galaxies, stars, planets, moons, comets, and atoms. As we move to smaller entities, such as quarks, the landscape becomes less clear. Other phenomena, such as dark energy and dark matter, remain mysterious.

Reductionism[1] proceeds until all object groups may, at some point, be reduced to single, discrete particles. These discrete particles are not composed of smaller units of matter; thus, the process of decomposition ends. This philosophy considers all such particles to be Universal Matter, denoted by the set {f}. All members of the set {f} constitute a field, f, of such matter.[3]

One may ask, "Where does this leave us?" or "What is the nature of this field?" Let us take a closer look.

We first examine the Standard Model (SM) to determine whether it has already identified the nature of this field. Ethan Siegel writes:

> "Today, our experiments show no evidence for compositeness to any of the Standard Model particles down to the absolute limits of what we've been able to probe. Based on experiments conducted at high-energy particle colliders, we can conclude that each of the Standard Model particles is truly fundamental, at least as far as we've been able to measure: down to this scale of 10^{-19} meters."[4]

The current measurable limit of 10^{-19} meters is extremely small, yet it allows experiments to probe many SM particles. Could the field f be composed of SM particles? This is possible if we consider the field to consist of either bosons or fermions; however, it may be more appropriate to view it as a new particle-based entity.

George F. R. Ellis suggests that "reductionism is best viewed as causation from both the top-down and the bottom-up."[5] It appears, then, that if we view the universe from a top-down perspective, the fundamental matter of the universe is described by the Standard Model. However, this natural philosophy also requires a bottom-up account of causation to counterbalance the top-down position. To begin this exercise, I contrast the largest nearby object known in physics, the Milky Way galaxy, with one of the smallest objects, a hydrogen atom.

The size of our galaxy, the Milky Way, remains a subject of debate due to the continual availability of new information. According to Pat Brennan at NASA, "our galaxy is about 100,000 light-years across,"[6] or approximately 10^{21} meters. The diameter of a hydrogen atom is twice its radius of 53 pm, or $1.06x10^{-10}$ meters.[7]

What, then, is the size of a particle that has the same proportional relationship to hydrogen as hydrogen has to our galaxy? Let us define a sizing factor for the relationship of hydrogen (h) to the galaxy (g) by setting:

$$g = 10^{21}, \quad h = 1.06x10^{-10}$$

The sizing factor (sf) is then given by:

$$sf = \frac{h}{g}$$

Thus, a bottom-up estimate for the size of a particle in {f} is:

$$h * sf \approx 1.124x10^{-41}$$

This is an extremely small value, well below the Planck length of $1.6x10^{-35}$ meters. However, as Siegel writes:

> "…You can't extrapolate what we know about even the small-scale Universe to arbitrarily small distance scales. If we decide to go down to below about 10^{-35} meters - the Planck distance scale - our conventional laws of physics only give nonsense for answers. Here's the story of why, below a certain length scale, we cannot say anything physically meaningful."[8]

Thus, 10^{-35} meters is not necessarily the smallest particle size that can exist. However, due to energy constraints and uncertainty in measurement technology, it represents the smallest length scale that can currently be meaningfully discussed within established physics.

In this discussion, the reader may struggle to maintain a sense of scale. To address this issue, we first list several well-known objects from astronomy and physics. We then compare their sizes relative to that of our galaxy and estimate the sizes of the composite particles discussed in this study. Table 1 summarizes these objects.

Object	Size in Meters
Galaxy	1.00E+21
Solar System	5.99E+12
Sun	1.39E+09
Average Planet	5.01E+07
Average Asteroid	1.50E+02
Space Dust	2.00E-06
Hydrogen	1.06E-10
Proton	8.60E-16
Electron	1.00E-14
Photon	1.00E-20
Quark	1.00E-18
Planck Length	1.62E-35

Table 1: Object Sizes in Meters

Next, as above, we select several of these objects, define size factors for them, and project sizes for the particles discussed in Table 2.

Drill		**Drill**	**Drill**
Down		**Down**	**Down**
Name	**Compare**	**Factor**	**Size**
{f}	Hydrogen	1.06E-31	1.12E-41
Level I	Asteroid	1.50E-19	1.59E-29
Level II	Planet	5.01E-14	5.31E-24
Level III	Sun	1.39E-12	1.47E-22
Prime	Solar System	5.99E-09	6.35E-19

Table 2: Projected Object Sizes

Note: The "Compare" column is always defined relative to the size of the galaxy listed in Table 1. For example, the drill-down factor for an individual member of the set {f} is $1.06x10^{-31}$. Multiplying this factor by the size of a hydrogen atom, $1.06x10^{-10}$ meters, yields an estimated size of $1.12x10^{-41}$ meters for a particle of {f}.

The drill-down factor for a Level I particle is $1.50x10^{-19}$. Multiplying this factor by the size of a hydrogen atom, yields an estimated size of $1.59x10^{-29}$ meters. Similar calculations apply to the remaining entries in Table 2.

With these calculations established, we are prepared to present the natural philosophy. We begin by describing the universal join process required to construct a hydrogen atom.

2 The Universal Join Process

Particle members of the set {f} are irregular in shape and size and are not evenly distributed within the field f. Perturbations in the field allow members of {f} to move and act independently of one another.

Any particle member of the set {f}, such as X, when in motion, may collide with another particle in the field f, such as Y. The collision produces both a complex result and a simple result. The complex result has two possible outcomes:

- Due to the irregular shapes of particles X and Y, X may collide with Y and remain joined, forming a single composite particle composed of two members of the set {f}. The collision may also introduce rotation in the resulting composite particle.

- Alternatively, X may collide with Y and then separate, remaining two discrete particles of {f}.

In either outcome, there is an additional simple result:

- The collision may cause small fragments of either X, Y, or both to break off from their parent particles. These fragments become separate subparticles of {f}, which may be labeled "sub-f" particles. They retain the properties of their parent particles, and energy is transferred from the parent particles as a result of the collision.

As newly formed sub-f particles are created, they become members of the set {f}. As such, they occupy spaces in and around their parent particles, forming a localized influence on neighboring particles and other sub-f particles.

The sole dimension of size permits collisions and potential joining of these particles, resulting in either new particles or larger composite particles. The collisions and joining of particles of {f}, sub-f particles, or composite particles impart a second dimension, rotation, to composite particles due to kinetic energy transferred during interaction.

Note One: The Universal Join Process clearly involves particle movement and energetic collisions. This philosophy does not yet address these issues, as the necessary objects and processes have not been fully defined. Particle movement and energy are treated at length in a separate paper and will be cross-referenced when posted to arXiv.

Note Two: The following discussion focuses on a direct path from individual particles of {f} to a hydrogen atom. This should not be construed as implying that this path exhausts all possibilities. The types of particles produced by the processes discussed here represent only a few, though critically important, among the myriad possibilities found in nature.

3 Composite Particles

3.1 General

The growth of a composite particle mirrors embryonic human development. In human growth, the embryo leaves its "ball of cells" stage after the fourth week of development, at which point cells begin to form the detailed anatomy of a baby[9].

In the previous section, a composite particle entered its primary stage of growth when a particle of {f} collided with another particle of {f} and remained together as a single composite particle. It then begins this "ball of cells" stage and must incorporate additional particle members of {f} to continue development.

Composite particles grow in size through collisions with other particle members of {f} or with other composite particles. Sub-f particles accumulate within and around composite particles as growth proceeds. At this stage, composite particle growth mirrors another organic building process, namely plant growth and branch development. As noted by Nakamasu and Higaki, "Various branch architectures are observed in living organisms including plants."[10] These branch architectures may also be referred to as growth and branch strategies. In the growth levels described below, the designations of "growth" and "branching" indicate the dominant developmental process at each stage.

For clarity and discussion, we set out three attained levels of composite particle growth:

- **Level 1** (CPL1) Growth. This is the initial growth phase in the formation of a composite particle (CP). Individual members of {f} physically join together, increasing the size of the forming particle.
- **Level 2** (CPL2) Branching. At this level, CPL1 particles join together. The key to growth at this second level is the relative sizes and directions of rotation of the colliding CPL1 particles. Two types of growth are observed in CPL2 interactions: series and quantitative.
- **Level 3** (CPL3) Growth. Finally, the two types of CPL2 particles join to form CPL3 particles. These CPL3 particles combine both series and quantitative structures into single hybrid particles.

3.2 Level 1 Growth

This level involves existing composite particles joining with other {f} particles, including sub-f particles, thereby increasing their size. The particles of {f} do not aggregate like a ball of string compacted without space between individual particles. Rather, the structure resembles a bird's nest of sticks, twigs, and plant matter, or a ball of dust and hair on a hard floor. Various existing particles connect and form stable interactions with new particles, but none of the particles are connected to all others.

Accordingly, space must be allowed between, within, and around the members of the forming CPL1 particle. Sub-f particles may also remain within the composite structure, adding internal space to the CPL1 particle. Approximately $1.5x10^{12}$ joins are required to create a fully mature CPL1 particle. As detailed in Table 2, the relative size of the CPL1 is approximately $1.6x10^{-29}$. This remains well below the measurement threshold of 10^{-19} meters[4], but the first phase of growth is complete.

As the CPL1 particle reaches maturity, it begins the branching phase. Collisions shift from interactions with other particles of {f} to interactions between mature CPL1 particles. Thus, the next stage of bottom-up reductionism occurs in the formation of CPL2 particles.

3.3 Level 2 Branch and Develop

As a CPL2 particle begins forming, development branches into two types of growth: series and quantitative. Given two CPL1 particles, A and B, such that A collides with B, two factors determine the resulting join:

- **Size**
  - Series joins are most effective when A and B are similarly sized.
  - Quantitative joins occur when one particle, for example A, is larger than B.
- **Direction of Rotation**
  - Series joins occur when A and B rotate in opposite directions.
  - Quantitative joins occur when A and B rotate in the same direction.

The following discussion examines particle rotation as it relates to the point of contact, or collision, between two particles. The directions of rotation are not absolute but are defined relative to the other particle at the moment of collision. At the point of contact, the particles may rotate in opposite directions, or their rotations may be the same.

Figure 1 and Figure 2 illustrate the effect of rotation at the point of contact, labeled "X." Figure 1 shows that when two particles rotate in opposing directions about their respective centers, they move in the same direction at the point of contact "X." Figure 2 shows the opposite case. When the rotations, denoted by "R," are in the same direction about their respective centers, the directions of motion at the point of contact "X" oppose each other.

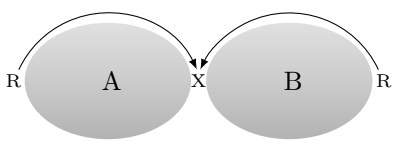


Figure 1: Particles Rotating in Opposite Directions

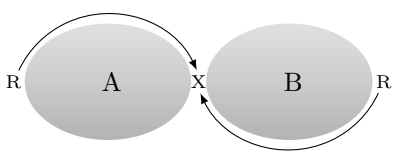


Figure 2: Particles Rotating in the Same Direction

3.3.1 Series Joins

In preparation for discussing this area of branching, I sought sources that explore the effects of media exchange, such as water, between two rotating bodies. Most articles focused on the effects of rotating bodies on the medium itself rather than on the effects of the medium between the two bodies. In addition, these discussions generally addressed on bodies moving through, or within the flow of, the medium. However, I did identify discussions of the Magnus effect.

In the present context, CPL1 particles rotate in, but do not move through, the field f. The flow of field particles around one particle, such as "A" in Figure 3, may nevertheless be considered to be in motion relative to "B" in the same figure, particularly in the region labeled "X."

Two CPL1 particles with opposing overall directions of rotation may begin forming a CPL2 particle. Figure 3 shows that although their overall directions of rotation oppose each other, at the point of contact "X" both particles rotate in the same direction.

The Magnus effect states, "On the side where the motion of the whirlpool is in the same direction as that of the direction of flow to which the object is exposed, the velocity will be increased."[11] Thus, the flow of field particles f at the point of contact "X" will increase. "Then, according to Bernoulli's principle, the pressure is lower on the side where the velocity is greater, and consequently there is an unbalanced force at right angles to the fluid flow."[11] As a result, the force "F" is directed toward each of particles A and B.

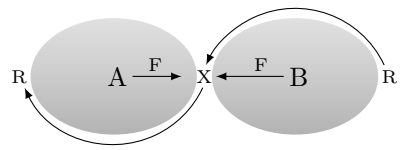


Figure 3: Series Join Particle Rotations

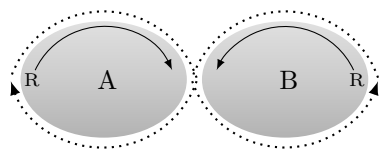


Figure 4: Series Join Particle Exchange

Series joins are most effective when the two CPL1 particles are approximately equal in size. The compatible direction of rotation at the point of contact, combined with the Magnus effect, allows particle A to couple with particle B, and vice versa, thereby forming a persistent series join.

Figure 4 shows that as A and B continue rotating against each other, the small {f} particle members of each CPL1 particle, which initially rotate around their respective parent particles, peel off and become entrained in the other particle's rotation. The exchange of {f} particles continues and forms an independent flow jointly maintained by A and B. This flow forms a figure-eight pattern and stabilizes the series join.

Additional series joins may occur when other CPL1 particles come into contact with an existing, forming CPL2 series particle. As illustrated in Figure 5, the direction of rotation of the newly interacting CPL1 particle, denoted C, must oppose the direction of rotation of series member B, where A and B already constitute a CPL2 series particle.

The small surface particles of the newly joined CPL1 particle C peel off and join the continuing figure-eight flow of the existing CPL2 series particle. As a result, the series join evolves into a lengthening figure-eight flow around and between all CPL1 members of the growing chain. In Figure 5, four additional CPL1 particles, C, D, E, and F, have joined the original members A and B.

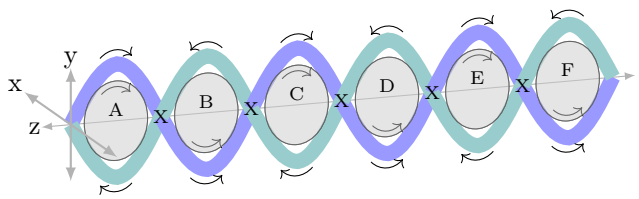


Figure 5: Series Particle Growth

Lattice Growth Figure 6 illustrates that growth may also occur at intersections along an axis. Any member of an existing CPL2 series particle may join with any member of itself or another CPL2 series particle. The only requirement is that, at the point of contact, the new CPL1 particle must be fully compatible with all rotational relationships in the series.

Extensive CPL2 series growth at many points along the axis of rotation allows the matrix to be viewed as a surface (Figure 6). With additional series joining, further surface area may be created, expanding the growing surface layer formed through the series join. Complex surfaces may result when a portion of an existing series join comes into contact with another area of itself and successfully completes the join process, or when the edge of one surface comes into contact with the edge of another and successfully forms a new series join.

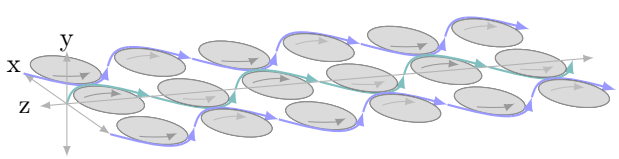


Figure 6: Layered Series Join On X-Axes

Conductance Complex series-joined surface areas may form gaps between matrix layers of a CPL2 particle. Small forming CPL1 particles may then come into proximity with the surface-layered edges of a CPL2 series join. The approaching particle may rotate in either the same or the opposite direction relative to the series join and may therefore participate as either a series or quantitative join with a CPL2 edge particle. The result differs from regular CPL2 joins because of the size difference between the approaching CPL1 particle and the established CPL2 series particle.

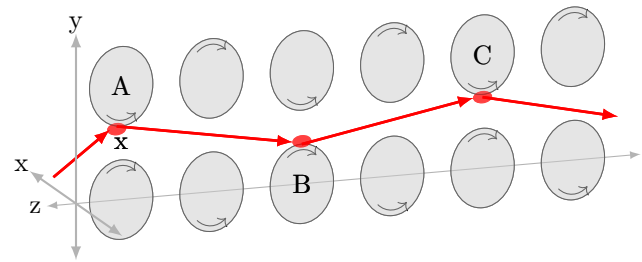


Figure 7: Series Layer Conductance

Figure 7 illustrates a CPL1 particle, x, approaching a CPL2 series particle layer. It collides with an existing member of the CPL2, particle A, which attempts to join with x based on their respective directions of rotation. The join may be successful; however, the focus here is on failed joins. Following collision, x may rebound inward in the direction of rotation of particle A and continue moving between the surface layers of the CPL2 particle.

Rather than rebounding along a path parallel to neighboring surface layers, x is likely to follow a nonparallel path and collide with another member of the CPL2 series particle. After colliding with A, x rebounds and collides with B, then rebounds in the direction of rotation of B. It subsequently collides with C and rebounds again, eventually exiting the series layer.

In this manner, x may be conducted through the CPL2 surface-layer gaps and exit at some point within, between, or at the far end of the series join surface layers. This conducted path may be conceptualized as a flow of matter through the CPL2 layers. Thus, CPL2 series particles introduce a third dimension to CPL2 growth: conductance.

3.3.2 Quantitative Joins

Series joins may be viewed as a process of accretion and coalescence, often understood as building single objects. As noted by La Salvia, "Through this process, called 'accretion,' these microscopic particles formed larger bodies that eventually became planetesimals with sizes up to a few kilometers across." [12] Quantitative joins align more closely with this expectation by producing growing single particles.

Quantitative joins involve collisions between particles of different sizes. Typically, one CPL1 particle, B, is larger than the other, A (Figure 8). At the moment of contact, A and B rotate in the same direction and begin forming a CPL2 quantitative particle.

Although their overall directions of rotation are the same, at the point of contact X they rotate in opposing directions. The contacting surfaces of A and B begin to accumulate constituent members against each other. A begins piling up matter above the contact point, and B below it (Figure 9).

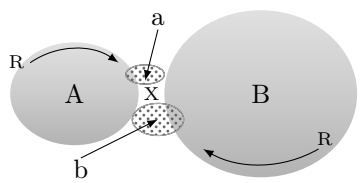


Figure 8: Conflicting Rotation, Buildup Begins

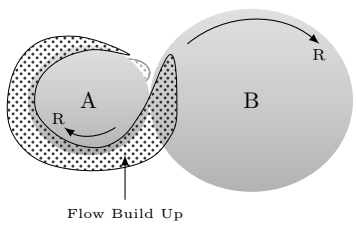


Figure 9: B Overflowing A

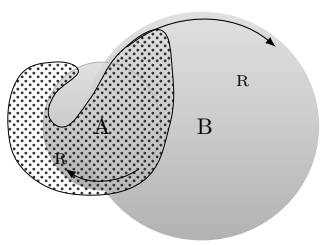


Figure 10: B Absorbs A

Because of their unequal sizes, the larger particle B accumulates a greater amount of constituent members of {f}, eventually overflowing particle A. By overflowing the smaller particle, B draws A into itself and absorbs it (Figure 10). The two thereby become a single quantitative CPL2 particle.

CPL2 quantitative particles either continue gathering CPL1 particles into themselves or are incorporated into other forming quantitative particles. In either case, their size increases.

After quantitative joining, the unified particle continues to rotate. Some overflow particles remain near, but detached from, the growing particle, forming a cloud of particles moving in the same direction as the primary particle's rotation.

Note: CPL2 joining is not always successful. Interacting particles may separate or remain weakly joined, allowing later interactions to break them apart. Collisions may also produce sub-f fragments. These fragments may later be assimilated in quantitative joins and accumulate in and around forming CPL2 particles.

3.4 Level 3 Growth

CPL2 growth, necessary for constructing a hydrogen atom, exhibits two branching strategies: series and quantitative. The sizing chart in Table 2 indicates that a mature CPL2 particle is approximately $5.3x10^{-24}$ meters in size. The number of joins required to reach this stage is estimated to range from 10^5 to 10^6.

Quantitatively joined particles may also assimilate CPL2 series particles. The joining process begins as in a regular quantitative join, with both particles rotating in the same direction. At the point of contact, opposing surface motions cause surface particles to accumulate against one another.

However, the matrix-like surfaces of the CPL2 series particle alter the joining dynamics. Because of its spread-out matrix structure, the series particle initially appears larger in volume, where size refers to volume rather than total constituent count. As the larger quantitative particle overflows the series particle, the latter wraps around and folds into it. The series particle is thereby integrated and contributes its conductive properties.

Through this process, CPL3 particles evolve into hybrid particles that combine the volumetric properties of quantitative particles with the conductance patterns of series particles. As CPL3 particles interact with other CPL2 particles, conductance becomes increasingly significant, acquiring importance equal to rotation and size.

4 Advanced Level 3 Growth

As additional CPL2 series particles are incorporated into CPL3 particles, conductive patterns emerge. These patterns arise from the CPL3 particle's ability to conduct matter flow into, through, and out of channels created during formation. Areas of inflow act as receivers, while areas of outflow act as emitters. These channels play a critical role in further maturation.

Channel patterns, along with sufficient size and mass, contribute to sorting CPL3 particles into advanced types. Some patterns may be generalized as either Effective or Ineffective CPL3 types. A third pattern, Complementary, may also emerge.

- **Effective:** Many channels successfully conduct particles into, through, and out of the CPL3 particle.
- **Ineffective:** Only a few channels successfully conduct matter into, through, or out of the CPL3 particle.
- **Complementary:** These particles exploit the properties of certain effective CPL3 particles and join at their midsections.

4.1 Effective Conductance

Effective CPL3 particles channel particulate matter into receivers and out of emitters. Each effective particle has two such emitter-receiver circuits, as shown in Figure 11. Conducted particles may flow out of an emitter and then return to a receiver. In the illustration, the effective particle rotates end over end, indicated as direction R, causing the conducted particles to flow backward, opposite to the direction of rotation, before returning to the receivers.

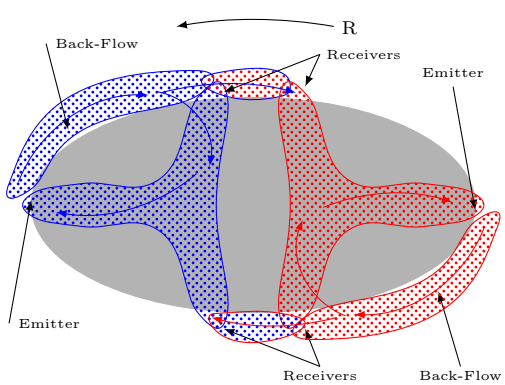


Figure 11: Effective Particle (Cross Section)

Not all conducted matter flowing from an emitter will be channeled back into a receiver. In addition, that particles in the neighborhood of the effective particle may enter the flow through one of the receivers, join existing particles in the loop, and flow toward an emitter.

4.2 Ineffective Conductance

Ineffective particles do not form many, if any, emitter-receiver circuits. Although areas within these particles can serve as receivers and emitters, they are arranged in a haphazard fashion and are not prone to creating conducted flow channels. As will be shown in related papers, however, ineffective particles are critical to the formation of regular matter.

4.3 Complementary Conductance

Complementary effective particles, abbreviated CmP, form by combining two effective particles. The particles join at their midpoints to form a plus shape. A representation of a CmP is shown in Figure 12. Here, the CmP is depicted as two ellipsoidal spheres composed of shaded frames. The axes and rotations of the ejected particle are as follows:

- End over end on the z-y plane, with rotation about the x-axis. The emitter end is denoted by a red dot.
- Around and around on the x-y plane, with rotation about the z-axis. The emitter end is denoted by a blue dot.

Figure 12 may be viewed as two clock faces aligned in three dimensions. The two clock faces bisect each other at (x, y, z) = (0, 0, 0). When viewed along the x-axis, the red effective particle's clock face appears on the z-y plane, with the red dot is at the 12 o'clock position. When viewed along the z-axis, the blue effective particle's clock face appears on the x-y plane, and the blue dot is likewise at the 12 o'clock position.

The rotation of each particle is relative to the other. For example, if the red particle rotates clockwise, the blue particle will also rotate, and vice versa. Clockwise rotation is neither more nor less effective than counter-clockwise rotation.

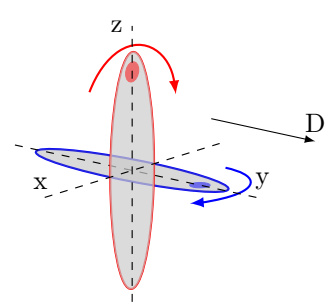


Figure 12: Complementary Particle Rotations

As the two particles approach one another, they rotate in the same direction and therefore begin with a quantitative join. Figure 13 illustrates how conducted matter flows through the CmP, shown as middle to left and middle to right. This structure is similar to the effective particle described in Figure 11, but contains only a simple channel through its center. Note the two illustrated top

and bottom sets of portals and how they are separated into several channels. These channels serve as receivers or emitters, depending on their specific makeup. As conducted particles flow through the CmP, some of the flow is routed into or out of the midsection of the CmP. The conducted inflow and outflow at the midsection form a conductive bond between the two constituent particles of the CmP.

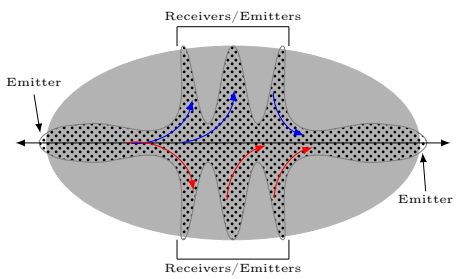


Figure 13: Complementary Particle Flow

As the CPL3 composite particle matures, the overall branch and grow strategy comes to fruition. CPL1 represented a growth stage. CPL2 branched and grew by using series and quantitative join processes. In the CPL3 stage, growth once again predominates and brings the particle to maturity.

5 Prime Particles

5.1 Status

The world of Level I through III composite particles is relatively straightforward to describe in terms of size. The primary contributor to growth has been the addition of other and/or smaller particles. As noted previously, review of Table 2 suggests that the projected size of a CPL3 particle is approximately $1.47x10^{-22}$, with a count of CPL2 units at about $2.5x10^{06}$. The count varies with different configurations and parameters.

Effective conducting particles tend to be somewhat smaller than ineffective conducting particles due to a greater presence of conductive channels formed during their construction. In sizing studies, effective particles measure approximately $9x10^{-23}$ meters, whereas an ineffective particle measures approximately $1x10^{-22}$ meters.

5.2 Primes

An mature CPL3 particle is labeled a Prime particle. This change in terminology does not indicate any substantive change in composition. Rather, it signifies that prime particles are prepared for a different join process. As suggested by Table 2, the prime particle measures approximately $6.4x10^{-19}$ meters and is composed of approximately 4,300 CPL3 units. Prime particles consist of either an effective or an ineffective particle, but not a complementary particle.

Prime particles are proposed to become quarks with further development, although this view diverges from current scientific understanding. As noted by Siegel, "…If you bombard electrons or quarks or gluons with high-energy particles, they show no evidence of internal structure at all, down to a size of 10^{-19} meters."[4]

The projected size of the prime particle is close to the estimated size of a quark, approximately $1.0x10^{-18}$ meters, as noted in Table 1. The remainder of the required size, and certainly its mass, would result from joins with other prime particles.

5.3 Paired Primes

An actively conducting effective prime particle, abbreviated Ep, may interact and join with an ineffective conductive prime particle, abbreviated Ip. The two primes rotate in the same direction, so the process begins as a quantitative join, and the two particles begin to accumulate constituent particles of {f} against each direction of rotation. However, the additional dimension of conductance modifies the standard of a quantitative join process.

At the point of contact, pools of conducted particles quickly develop. One pool consists of conducted particles from the Ep, and the other consists of the conducted particles from the Ip. Direction R, as denoted in Figure 14, represents the direction of relative rotation at the point of contact. While either prime particle could exploit emitter or receiver channels in the other, the Ep, as illustrated in Figure 14, is much better positioned to exploit channel flows into the Ip.

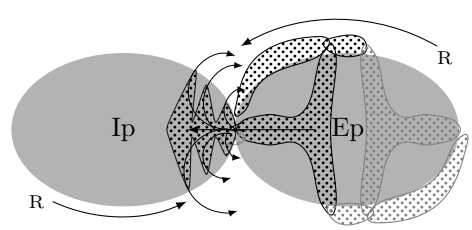


Figure 14: Ep-Ip Paired Prime

Figure 14 illustrates that an Ep and an Ip may successfully join if the Ip possesses a sufficient number of emitter-receiver channels and the Ep is properly positioned to emit particles into the Ip's receiver channels. The Ep may emit particles into one of the Ip's receiver channels, shown at the Ip's 3 o'clock position, which

then flow into and out of the Ip's emitter channels located above and below that position. Some of these outflow particles may then be captured in the Ep's backflow of conducted particles and subsequently flow into the Ep's receiver circuit at the 12 and/or 6 o'clock positions. If a sufficient number of these receiver-emitter flow loops form between the two primes, they may persistently join as a single paired prime particle. Once the join is complete, the join, the paired prime rotates overall in the direction of the Ep.

At this stage, the effective and ineffective members of the paired prime are compatible and do not intrinsically produce rotational acceleration. The stress on conducted particles in a successfully paired prime is nominal and stable. These conducted particles represent the initial presence of the strong force, "which originates in a property known as colour" …and… "within its short range, about 10^{-15} meter, roughly the diameter of a proton or a neutron, the strong force appears to become stronger with distance, unlike the other forces."[13] Given the size of a prime noted above, this interaction occurs well within that range.

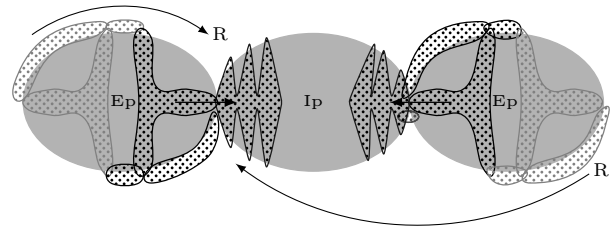
Success is not guaranteed. An Ep may fail to find a sufficient number of receiver channels in the Ip; indeed, the Ip may have none. Without adequate flow into each particle's channels, or without sufficient flow occurring in a timely manner, the join effort will fail and the particles will separate. Some transfer of conducted particles may occur, and groups of matter may be broken off or lost in the collision. The joining effort may also produce sub-f particles.

5.4 Triple Primes

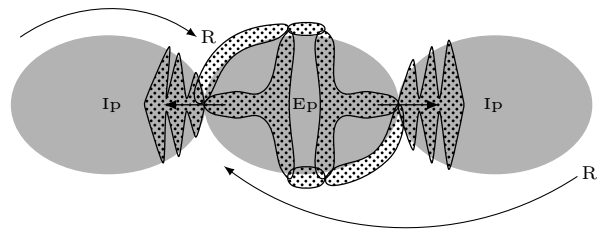
Paired primes serve as building blocks that may join with a third prime particle for continued development. The third prime joins with only one member of the paired prime, not both. The process begins as a quantitative join. A triple prime may form in one of two ways:

- **Active:** The ineffective prime of a paired prime joins with an effective prime (see Figure 15a).
- **Passive:** The effective prime of a paired prime joins with an ineffective prime (see Figure 15b).

Given a paired prime notated as Ep-Ip, four possible outcomes exist:



(a) EIE Triple Prime



(b) IEI Triple Prime

Figure 15: Triple Primes

- **The Ep of an Ep-Ip pair attempts to join with an Ep: failure.** Strongly conducted flows oppose and rebuff one other.
- **The Ep of an Ep-Ip pair attempts to join with an Ip: success.** Figure 15b. The resulting triple prime is Ip-Ep-Ip, abbreviated IEI.
- **The Ip of an Ep-Ip pair attempts to join with an Ip: failure.** Weak or nonexistent conducted flows do not establish stable channels.
- **The Ip of an Ep-Ip pair attempts to join with an Ep: success.** Figure 15a. The resulting triple prime is Ep-Ip-Ep, abbreviated EIE.

5.5 The Conductive Strong Bond

The conductive joins between combinations of Ep and Ip, whether as paired or triple primes, constitute the substantial basis of the strong force between prime particles. As described in Encyclopaedia Britannica:

> According to quantum field theory, particles can be held together by a "charge-exchange" force, which is carried by charged intermediary particles. Heisenberg's application of this theory gave birth to the idea that the proton and neutron were charged and neutral versions of the same particle-an idea that seemed to be supported by the fact that the two particles have almost equal masses. Heisenberg proposed that a proton, for example, could emit a positively charged particle that was then absorbed by a neutron; the proton thus became a neutron, and vice versa. The nucleus was no longer viewed

> as a collection of two kinds of immutable billiard balls but rather as a continuously changing collection of protons and neutrons that were bound together by the exchange particles flitting between them. [13]

At the level of triple primes, emitted and received conducted particles may be viewed as charged, flowing from an effective particle to an ineffective particle and vice versa. As observed in the quotation, protons and neutrons are similar in size. Likewise, effective and ineffective primes are also similar in size.

These joins experience substantial stress while keeping the triple prime particles attached. The stress may be likened to springs that are continuously compressed, extended, and twisted. The imposed load amounts to stored kinetic energy and contributes to the mass of the triple prime through rotational acceleration.

The conductive joins of the CPL1 withstand stress because of their construction. No single particle of {f}, nor any subgroup of {f} particles, is directly connected to all other constituent particles in a CPL1 particle. The assembly processes described in this philosophy produce loosely connected structures in which constituent particles form coherent larger particles through partial connectivity. Stress on a conductive join exploits this structural property.

As conductive joins are stressed by larger joined prime particles, conducted particles are stretched longitudinally and thinned. This permits additional conducted particles to enter the flow channels, where they are likewise stretched and thinned. The result is increased joint strength and greater stored kinetic energy.

6 A CmP Joins a Triple Prime

Both active and passive triple primes may attempt to join with new effective or ineffective primes. Of particular interest is a join between an EIE triple and a complementary particle, CmP. When the CmP contacts the flow shell of an EIE triple prime, and both rotate in the same direction, the collision initiates a quantitative join. The EIE triple quickly overflows the CmP. However, the result may differ from the classic scenario in which the CmP is absorbed, because the paired nature of the CmP may alter the outcome.

6.1 CmP Join

The CmP now resides within the conductive flow of the EIE. Figure 16 illustrates the relationship between the CmP and the flow. The z-axis represents distance from the center of the EIE. The EIE's conductive flow is represented by a light gray field of dots extending along the z-axis, with the top of the field farthest from the EIE particle. The field rotates in direction R on the x-axis around the center of the EIE triple prime. Each particle of the CmP, red and blue, occupies its respective 12 o'clock position. In this illustration, the red particle rotates parallel to the field, and the blue particle rotates perpendicular to it.

The CmP tumbles along the x-axis, engulfed by the conducted flow of the EIE. One of the pair, for example the red particle, may align with the flow direction. Its receiver end begins taking in conducted EIE flow, and its emitter end emits that flow back into the main stream. While the red particle remains anchored in the primary flow direction, the blue particle acquires sufficient flow to construct its own conducted flow circuit perpendicular to that of the red particle. The CmP pair continues their receiver-emitter circuits as they tumble around the EIE.

6.2 Electromagnetism

As the CmP tumbles around the EIE, its red and blue particles emit patterned conducted flow. In this example, the blue particle emits conducted matter away from the EIE and then toward it in a wave pattern on the z-y axis. The red particle emits a similar wave pattern on the x-y axis. For an observer looking along the x-axis from left to right, when the blue particle's conducted matter flows upward and away from the EIE, the red particle's conducted matter flows toward the right and away from the z-axis. In this manner, the CmP generates a dual wave pattern analogous to electromagnetism.

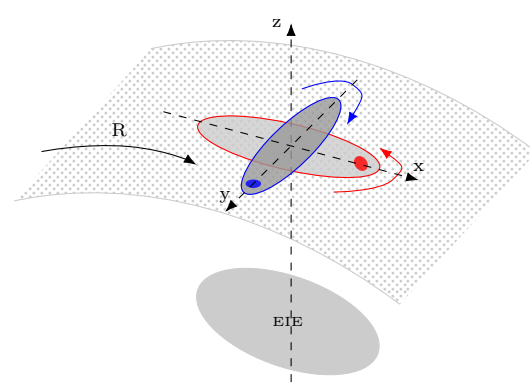


Figure 16: Triple Prime with Complementary Particle

6.3 Quantitative Pull

The conducted flow of EIE matter through the CmP enables it to resist the quantitative pull of the EIE triple particle. The CmP's emitted flows move outward into the EIE stream, and some portion is directed inward toward the EIE triple prime. The EIE emitters push back

against the CmP emitters, creating pressure that drives the CmP outward.

Sub-particles of {f} assist in this resistance. These energetic sub-particles remain in and around the EIE and exert pressure in all directions. This action serves to:

- Add space within the triple prime, maintaining structural openness and facilitating internal repair.
- Add loft to the triple prime's conductive flow, preventing it from succumbing to the quantitative pull of the structure as a whole.

At a critical point, the CmP escapes the quantitative pull because of the conducted EIE matter flowing through it. The energy of the sub-particles acts as a spring, propelling the CmP outward along the larger flow pattern.

The CmP nevertheless remains within the conductive flow stream of the EIE triple particle, anchored by its receiver-emitter flows. The EIE continues its quantitative joining effort, repeatedly attempting to draw the CmP inward. These efforts remain persistent but unresolved, and the CmP continues tumbling in the EIE's conductive flow.

6.4 Hydrogen

The CmP and the EIE together form a stable atomic particle: hydrogen. The EIE triple particle, with effective prime particles on either side an ineffective prime particle, forms the proton nucleus. The complementary particle, continually tumbling within the EIE's conducted flow, functions as the electron.

6.5 Elements of the Periodic Table

The process of building the elements does not stop with hydrogen. Additional reading may be found in a public repository entitled "The Structure of Elements."[14]

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